

Assignment 3 | Bridge Pattern

Option A - Drawing / English reference report

Student: [YOUR_NAME] **Group:** [YOUR_GROUP]

Repository: [YOUR_GITHUB_REPOSITORY_URL]

Version: final local commit in LOCAL_COMMIT.txt; repository not published.

Study example. Section 9 requires students to write and understand their own work and limits AI use to reference examples and clarification under the syllabus. [A]

1. Two dimensions, one bridge

The shape hierarchy answers **what is drawn**: Circle retains a radius and Square retains a side. The renderer hierarchy answers **how it is described**: VECTOR, RASTER, or ASCII. Shape holds a Renderer-interface reference supplied through its constructor. Its subclasses delegate through that reference, rather than construct renderers or test concrete types. [A, sections 2-3]

Bridge fits because combinations are assembled using objects instead of one subclass per shape/rendering combination. Runtime replacement changes the renderer reference, not the shape. The program is an allowed text simulation: demo radius 2 and side 3, with String results. [A, section 2; P]

2. Five annotated Clean Code excerpts

1 / Separation of responsibilities

src/Circle.java:17-19

```
public String execute() {
    return getImplementation().renderCircle(radius);
}
```

Circle chooses the circle operation and supplies domain data. Renderer owns the rendering description, so backend details do not leak into Circle.

2 / Meaningful names

src/Renderer.java:2-3

```
String renderCircle(int radius);
String renderSquare(int side);
```

Operation and parameter names identify both the shape and its dimension; there are no unexplained generic action names.

3 / Small, focused methods

src/Square.java:12-14

```
public int getSide() {
    return side;
}
```

This getter only exposes the side value. It performs no rendering, printing, or state modification.

4 / Shared checking workflow

src/Main.java:69-70

```
boolean success = expected.equals(actual);
printCheck(testId, success, classes + " | result=" + actual);
```

checkResult is reused by T1-T4, T6, and T7. printCheck is also used by T5, keeping status printing and counters in one place.

5 / Encapsulated state

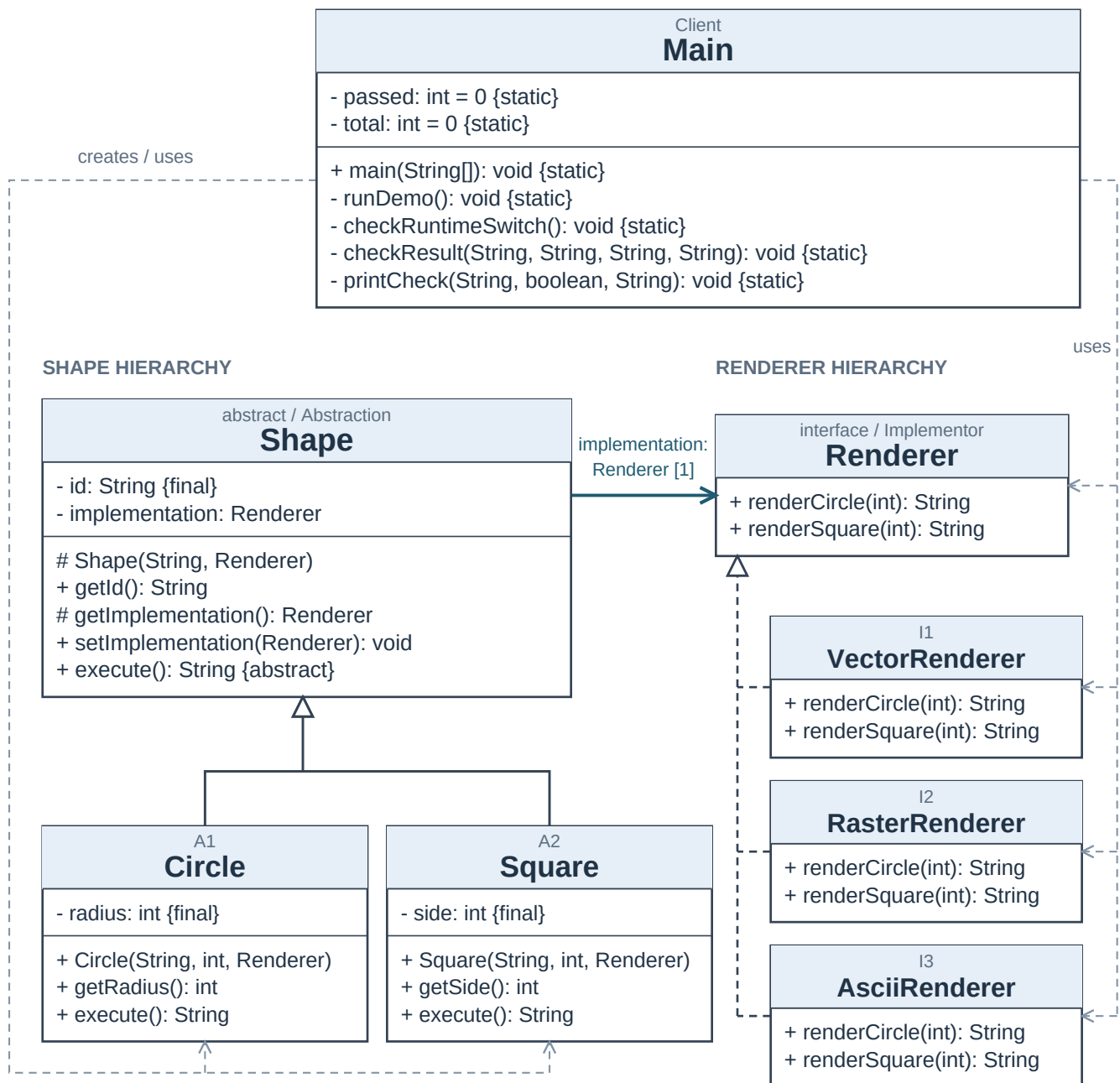
src/Shape.java:4-5; src/Circle.java:2

```
private final String id;
private Renderer implementation;
private final int radius;
```

The first two fields belong to Shape; radius belongs to Circle. ID and dimensions cannot be reassigned. The renderer stays replaceable only through the public null-checking setter.

3. UML of the final code

Both hierarchies, all three implementors, and the client. Method signatures use parameter types for compactness; all eight Java types are in the default package.



Notation. Hollow triangular arrows indicate inheritance or interface realization; dashed open arrows show client dependencies. The solid link labelled **implementation: Renderer [1]** is the stored non-null interface reference. + public, - private, # protected. Renderers may be shared across shapes, so exclusive lifecycle ownership is not asserted.

One delegated call

```

Main.main -> runDemo -> Circle.execute
-> Shape.getImplementation -> VectorRenderer.renderCircle(2)
<- "VECTOR circle radius=2"
-> checkResult -> printCheck
  
```

Constructors initialize the base part of the same Circle; they do not create a separate Shape object. execute returns the renderer result, and Main performs the comparison and printing. The standalone `uml.svg` can be enlarged without raster loss.

4. Checks and independent extension

Captured final result: **SUMMARY: 7/7 PASS**. The base version also ran successfully with T1-T5: **SUMMARY: 5/5 PASS**. Full console evidence is in demo-output.txt and base-demo-output.txt.

Check	Actual result / observed state
T1	VECTOR circle radius=2
T2	RASTER circle radius=2
T3	VECTOR square side=3
T4	RASTER square side=3
T5	sameObject=true; stateUnchanged=true ID: circle-switch -> circle-switch; radius: 2 -> 2 before: VECTOR circle radius=2; after: RASTER circle radius=2
T6	ASCII circle radius=2
T7	ASCII square side=3

T5 saves original, assigns current = original, switches only the renderer field, and evaluates original == current. It additionally compares ID/radius and both actual results. Status labels and the final count are computed, not fixed. [A, section 5]

Actual local extension evidence

Base: 8a4877597e1589f1293d2fa1f4c78ea0ca725b5a
I3 extension: a58562b5ab3fd11f84149343ff2a7ece0efd77e6

The base was committed after the I1/I2 checks. The next source commit added AsciiRenderer and updated Main for T6/T7. sources.txt was also updated. All existing abstraction classes, Renderer, I1, and I2 remained unchanged. The following committed source diff is archived as extension.diff: [A, section 4; G]

```
git diff 8a4877597e1589f1293d2fa1f4c78ea0ca725b5a HEAD -- src
```

```
A src/AsciiRenderer.java
M src/Main.java
```

Bridge, Adapter, and the cost of this design

Intent. Bridge separates two varying parts of a design. Adapter presents the required interface over an otherwise incompatible API. Here every renderer already implements Renderer; no legacy-interface translation is needed. [P]

Trade-off. This example introduces more classes and an interface call than a single-function simulation. Its explicit renderCircle/renderSquare contract is easy to trace, but a new primitive such as a triangle would require expanding the contract. Adding another renderer for the existing operations does not.

Verification scope. Built with javac 21.0.11 using --release 17 and run on OpenJDK 21.0.11, not separately on JDK 17. A temporary broken-renderer check produced FAIL and continued through T7; the shipped source remains the passing version. See VERIFICATION.md.

References and completion fields

- [A] Astana IT University. *Assignment 3 | Bridge Pattern*, supplied PDF, sections 1-9.
- [P] Apache Software Foundation. *Apache Calcite API: Glossary*, BRIDGE_PATTERN and ADAPTER_PATTERN.
- [G] Git project. *git-diff documentation*.
- [J] Oracle. *Java SE 17 javac and java command manuals*.

Before an actual submission, fill in student/group/repository details, update the report, and provide the real submitted repository URL and commit hash in Moodle. The local example history is not a published student repository. [A, section 6]